Artificial Intelligence in Education and Schools

Abstract : With the increase in studies about artificial intelligence (AI) in the educational field, many scholars in the field believe that the role of teachers, school and leaders in education will change. In this regard, the purpose of this study is to examine what possible scenarios are there with the arrival of AI in education and what kind of implications it can reveal for future of schools. The research was designed as a phenomenological study, a qualitative research method, in which the opinions of participants from different sectors were examined. The results show that schools and teachers will have new products, benefits and also face drawbacks with the arrival of AI in education. The findings point out some suggestions for use of AI and prevention of possible problems. While participants generally seem to have positive perceptions towards AI, there are also certain drawbacks, especially highlighted by teachers and academicians, regarding the future of teaching. Lawyers and jurists tend to focus more on legal grounds for AI in education and future problems, while engineers see AI as a tool to bring quality and benefit for all in the education sector. The ongoing digitalization, and especially the use of Artificial Intelligence (AI) in education, is opening up new learning opportunities. Education can become more individualized and less limited by time and place. ChatGPT illustrates how dynamic the developments in AI are. The growing interest in AI in general and AI in education benefits the necessary exchange between key stakeholders. This discussion paper presents three future scenarios for the use of AI in institutional education: higher education, continuing education, and schools. The focus is on overarching questions regarding the prerequisites and success factors for effective application, as well as identifying joint fields of action for key stakeholders. Five fields of action for the use of AI in institutional education are highlighted based on the outlined future scenarios: (1) interdisciplinary collaboration, (2) qualification programs and competence development, (3) digital infrastructure and personnel, (4) ethics and data sovereignty, and (5) data interoperability in educational contexts. The aim of this discussion paper is to stimulate cooperative exchange between relevant stakeholders.

Introduction : Artificial intelligence (AI), generally expressed by the general public as the ability of machines or computers to think and act as humans do, represents the efforts towards computerized systems to imitate the human mind and actions (Wartman & Combs, 2018). In this respect, the basic definition of artificial intelligence can be expressed as the skilful imitation of human behaviour or mind by tools or programs (Mohammed & Watson, 2019). According to Timms (2016), it may be an illusion of the current structure to think that artificial intelligence will come within the computer format used at home. It could get into our lives within different functions and shapes. Ng (2017) claims artificial intelligence to be the new electricity of this age. Artificial intelligence is a candidate to be presented as the basic building block of the Fifth Industrial Revolution by providing itself to be a powerful factor in ensuring economic development with its potential (see, Golic, 2019). That could be why investments in artificial intelligence broke a record in China with $40 billion in 2017 (Mou, 2019). In line with its earnings from AI, China is expected to increase its gross domestic product (GDP) by 26% ($7 trillion) by 2030. North America is expected to have a 14.5% increase ($ 3.7 trillion) in the same timeframe (PwC, 2017). These data make the added value and global impact of artificial intelligence more understandable for the future economy, and in our case, for the future of education, which in turn, directs the economy and workforce, paving the way for the new Industrial Revolution. The in-depth development of artificial intelligence will affect many

situations, from the restructuring of the social order in the broadest sense to the education and administration processes in classes and schools. Schools that are expected to adapt to the digital age and embed 21st century skills in their main agendas are some of the main institutions that could be most affected by the development of artificial intelligence. Karsenti (2019) points out that new forms of technology will fill in our lives and captivate our youth, and this case may leave schools with no choice but to make room for them. In this regard, how the stakeholders from law, business, education, and engineering perceive this development, and how they foresee artificial intelligence in regard to education form the focus of this study. Thus, the purpose of this study is to examine what the use of artificial intelligence in education means and what kind of implication it can reveal for the future of education, according to the opinions of the participants from different sectors.

COM 1: Ethical Issues in the Use of Generative Artificial Intelligence in Teaching and Research in Côte d'Ivoire

AKREGBOU Boua Paulin Sylvain
Peleforo GON COULIBALY University (UPGC), BP 1328, Korhogo, Côte d'Ivoire
ORCID Link | Phone: 07 09 10 40 56 | Email: sylvain.akregbou@upgc.edu.ci
Sub-theme: 1. Ethics of Artificial Intelligence in Education

Abstract
This study focuses on the ethical challenges of using generative artificial intelligence (GAI) in higher education and research in Côte d'Ivoire. Its relevance lies in the fact that this new technology has sparked major pedagogical and scientific interest in recent years in the education sector (Bouchereau & Roxin, 2022; Collin & Marceau, 2022). Several studies highlight practical uses for teachers and students, emphasizing the enormous potential of GAI in higher education (Chenelat, 2023; UNESCO, 2019).
The applications of this technology appear to be redefining the ecosystem of education and research worldwide, particularly in Africa. In Côte d'Ivoire, GAI is revolutionizing research methods and teaching practices (Doh-Djanhoundy, 2019; Koudou, 2023; Touré & Traoré, 2023).
In many sectors, this new technology continues to fascinate and intrigue. However, its introduction into higher education raises significant ethical concerns:

- What are the ethical and critical issues of GAI in higher education and research in Côte d'Ivoire?
- How exactly are teachers and students using this new technology?
- What is the contribution of the learner, teacher, and researcher in GAI-generated content?
- What about the academic and intellectual integrity of students, teachers, and researchers?

The method used for this exploratory study is qualitative, based on a socio-technical approach, utilizing documentary research and semi-structured interviews with teachers, researchers, and students. The analysis of the collected data will provide a better understanding of this evolving technology and, most importantly, its use in the education sector.
Through this work, we aim to understand the real ethical challenges of using GAI in higher

education and research in Côte d'Ivoire, exploring usage situations of teachers and students in their activities, how they appropriate GAI-generated content, and grasping their responsibilities.

Keywords: Generative Artificial Intelligence; ethical issues; usage; teaching; research.

International Scientific Conference AfricaDigitalEdu 2024
Theme: Artificial Intelligence and the Digital Futures of Learning in Higher Education
Date: 17–18 October 2024, Abidjan, Côte d'Ivoire

COM 2: Towards an Ethical Adoption of Artificial Intelligence in Education in Africa: Challenges and Opportunities

Dr. BOUFENARA KARIMA[1]; Prof. SEMICHI WIDAD[2]
[1] [ORCID NO: 0000-0003-3872-4875] | Phone: 0213550065005 | Email: karima.boufenara@univ-constantine3.dz
Faculty of Architecture and Urbanism, Constantine3 University, New City Ali Mendjeli, Constantine, Algeria.
[2] Faculty of Information Sciences, Constantine3 University, New City Ali Mendjeli, Constantine, Algeria.

Sub-theme: 1. Ethics of Artificial Intelligence in Education

Abstract
Artificial Intelligence (AI) in educational systems can transform teaching and learning methods by offering powerful tools to personalize learning paths and improve outcomes. However, this adoption raises ethical questions: Can it reduce social inequalities, or might it worsen them, particularly in Africa, where technological and socio-economic disparities are significant? This communication explores these issues, emphasizing the importance of ethical and responsible adoption.

We will discuss the benefits of AI in education, such as data analysis to personalize learning, the use of chatbots to support students, and automated assessment systems. Programs like *M-Shule* in Kenya and the *Kolibri* platform deployed in Africa offer individualized and autonomous learning opportunities. These technologies provide solutions to major educational challenges in Africa, such as the shortage of qualified teachers and limited access to teaching resources.

However, the use of AI in education also carries risks. Among the ethical challenges are data privacy, algorithmic bias, and the potential impact on teacher employment. The massive collection of students' personal data raises concerns about privacy. Algorithmic biases can exacerbate inequalities if systems are not fairly designed. Additionally, the automation of educational tasks could threaten teaching jobs, particularly in regions where the labor market is fragile.

To address these challenges, we propose an ethical framework for AI adoption in African education systems, based on four essential principles: transparency, fairness, responsibility, and participation.

- **Transparency** means that AI decision-making processes must be understandable to users.
- **Fairness** demands that AI systems must not discriminate and should contribute to reducing inequalities.
- **Responsibility** implies that designers and users of AI must be held accountable for their impact on education.
- **Participation** requires the active involvement of all stakeholders in the design and implementation of AI systems.
- **2.1.2 Future Scenario**
  Scenario for Educational Management and Implementation
  Professors currently spend a lot of time preparing, uploading, and adapting content for students. Future AI-based learning management systems (LMS) will suggest existing tutorials to professors during semester preparation based on course descriptions and learning objectives or competencies to be acquired. These tutorials can then be integrated into in-person teaching, supported and accompanied by digital tools. Natural language processing will be used to link various tutorials: by analyzing terms in course titles and descriptions, parallels to existing tutorials and courses can be identified, allowing cross-references to be created. Tutorials can be flexibly integrated into physical classes and adapted or expanded with the help of adaptive templates. Students can freely choose the language of the tutorials, which will be translated in real-time.
  This approach enables teaching to focus more on information not covered by existing tutorials, allowing the specific topics of a course to be better contextualized or deepened. Scripts for tutorials will be automatically generated. Competencies, skills, abilities, and strengths will be continuously analyzed based on individual profiles, enabling systematic curriculum planning and management across the institution. Exams, teaching content, and resulting study workloads will be personalized, estimated, and projected. Learning and performance types will be considered, influencing deadlines, learning formats, and methods suggested to students by the system. The system will actively explain to students which learning style or strategy is being followed to enhance their performance.
  To protect students' privacy, they will be able to decide which data are anonymized or personalized for access by instructors. Instructors can create analyses at any time, even without knowing the identities of the individuals involved unless consent is given. This facilitates the deepening of trustful relationships, such as mentoring. The learning situation will be designed collaboratively between professors and educational experts and will be implemented by lecturers, tutors, and professors. Both professors and students will be accompanied throughout the semester by a mentor.

- **Scenario for Learning and Assessment**
  Learning and assessment will become personalized and machine-supported in the future.
  AI-based systems will provide feedback during learning on the learner's performance status and suggest next steps (e.g., "Take a break!", "Watch video XY!", "Talk to peers!", "Book a coaching session!").
  Performance status will be captured via emotional expressions in the face and eye movements. These suggestions aim to help students sustainably reflect on their self-

regulation processes.
Tests, exams, and quizzes will be automatically generated and graded using neural networks from images and texts.
Chatbots will provide real-time hints about errors (e.g., spelling), formulation suggestions (like ChatGPT), illustration proposals, alternative calculation methods, or improved solutions (e.g., for legal case studies).
Particularly risky or dangerous learning situations will be simulated through virtual realities (e.g., chemical reactors, conflict resolution, de-escalation techniques, medical treatment training).
Careers will continue to be shaped by individuals, with AI systems merely offering suggestions.
Purpose, vision, goals, and interests will be tracked by the system and used as the basis for learning content and coaching offered by bots and instructors.
Self-awareness gaps will be addressed through communication analyses in groups, providing insights into strengths, dynamics, values, and behaviors.
Evaluations will no longer be limited to instructors but will be extended to include feedback from mentors, coaches, and peers, providing a 360-degree view.
In addition to learning future skills, personal development and alignment with individual life goals will be a primary focus.
Learners will move in intelligent spaces that adapt to their emotions and body temperature. Flow states will be identified through heart rate monitoring and protected from interruptions (e.g., automated phone silencing and “Do Not Disturb” signs on doors, possibly locking doors).
Motivation and performance levels will be reported back and positively reinforced through voice assistants or chatbots.

- Data collection and analysis from students and professors will be minimized to conserve energy.
  Only renewable energy and server waste heat will be used.
  The focus will be on skill acquisition and performance improvement.
  Data quality and structure will be continually optimized through the involvement of individuals and machine learning processes.
  For instance, instructors will review evaluations and correct mistranslations, while students will provide feedback on group composition and matching.
  Diversity and bias officers will ensure the quality of the AI system analyses.
  Only situations with potential risks to humans or the planet will be simulated in VR/AR environments, thus preserving the ethics and sustainability of the approach.
  Coaches and mentors will be carefully selected and trained over a longer process to ensure their attitudes and intentions align with the program.

3. Current Status of Information Literacy Education at Konan University

At Konan University, information literacy education is positioned within general education as "Information Practice A and B." These are mandatory courses for first-year students. In these classes, students gain practical skills such as basic usage of computers, the Internet, and Office applications, which are necessary for research activities and employment. In addition, they acquire practical knowledge and skills for using the latest information communication technology.

For example, in the "Information Practice A" course offered by the Center for General Education, the following topics are covered:

- Basic operations of PCs and networks
- Email and Internet research
- Word processing
- Spreadsheet basics
- Presentation software basics

In "Information Practice B," more advanced topics are covered:

- Data analysis using spreadsheets
- Creation of business documents
- Practical use of presentation software
- Ethical considerations and risks in information society

The courses focus on practical skill development necessary for students' academic life and later careers. However, education in Mathematics, Data Science, and AI is not yet included in a substantial form.

Adding Mathematics, Data Science, and AI content to this existing structure would require curriculum reform and retraining of instructors.
Also, given that first-year students already have full schedules with required subjects, simply adding new classes is not realistic.
Instead, modularization, offering online lectures, or using flipped classroom models are being considered as ways to implement this education.

4. Possibilities for Incorporating "Mathematics, Data Science, and AI" into General Education

Considering the current situation at Konan University, here are some possibilities for incorporating "Mathematics, Data Science, and AI" education into general education:

1. **Small Step Introduction:**
   Rather than starting with specialized content, begin with basic concepts like understanding data, recognizing the importance of data-based decision-making, and cultivating critical thinking about AI.

2. **Collaboration Across Faculties:**
   Instead of leaving everything to information-related faculties, promote interdisciplinary teaching across departments such as Economics, Sociology, Business, and Science.

3. **Utilization of External Resources:**
   Use MOOCs (Massive Open Online Courses) and certified online content to supplement areas where internal resources are insufficient.

4. **Project-Based Learning:**
   Incorporate elements of Mathematics, Data Science, and AI into practical projects and research assignments across various subjects.

5. **Stepwise Development of Specialized Courses:** Gradually prepare pathways for interested students to advance from introductory courses to specialized lectures in AI, data science, and related fields.

Additionally, to ensure that AI literacy education does not become superficial, it is necessary to design courses that encourage students to apply what they have learned to real-world problem-solving and communication skills.

5. Conclusion

The "AI Strategy 2019" sets an ambitious goal: for all citizens to acquire a new literacy in Mathematics, Data Science, and AI.
In universities, general education must evolve significantly to meet these expectations.

At Konan University, where information literacy education is already part of the curriculum, it is essential to consider ways to incorporate Mathematics, Data Science, and AI education within existing structures.
However, due to human resource shortages and time constraints, a gradual and flexible approach is necessary, making good use of interdisciplinary cooperation and external resources.

As society increasingly requires AI literacy, universities must not only impart technical knowledge but also foster the critical thinking needed to judge and apply AI outcomes appropriately.

The 21st century, with its technological revolution, has witnessed the rise of Artificial Intelligence (AI) as a transformative force, especially in the field of education. In Ecuador, this phenomenon has led to significant changes in secondary education, driven by the adoption of AI-based tools. These innovations, which include adaptive learning systems and virtual assistants, are redefining how teaching is delivered, paving the way for more personalized and efficient methods. However, this transformation process is not without challenges, such as the need to ensure equitable access to technology and to adapt educational policies to new digital realities.

AI in Ecuadorian secondary education promises significant growth in the quality of teaching. Technology allows for the automation of up to 40% of teaching tasks and offers improvements in key areas such as personalized learning, adaptive assessments, and intelligent tutoring. The evolution of online education, with the incorporation of intelligent and adaptable systems, has further enriched the educational experience. Virtual reality, for example, enables global classrooms and connects students in virtual environments. AI, as an interdisciplinary field, benefits from contributions from areas such as cognitive psychology and neuroscience, resulting in more effective implementation in education and ongoing research to maximize its usefulness.